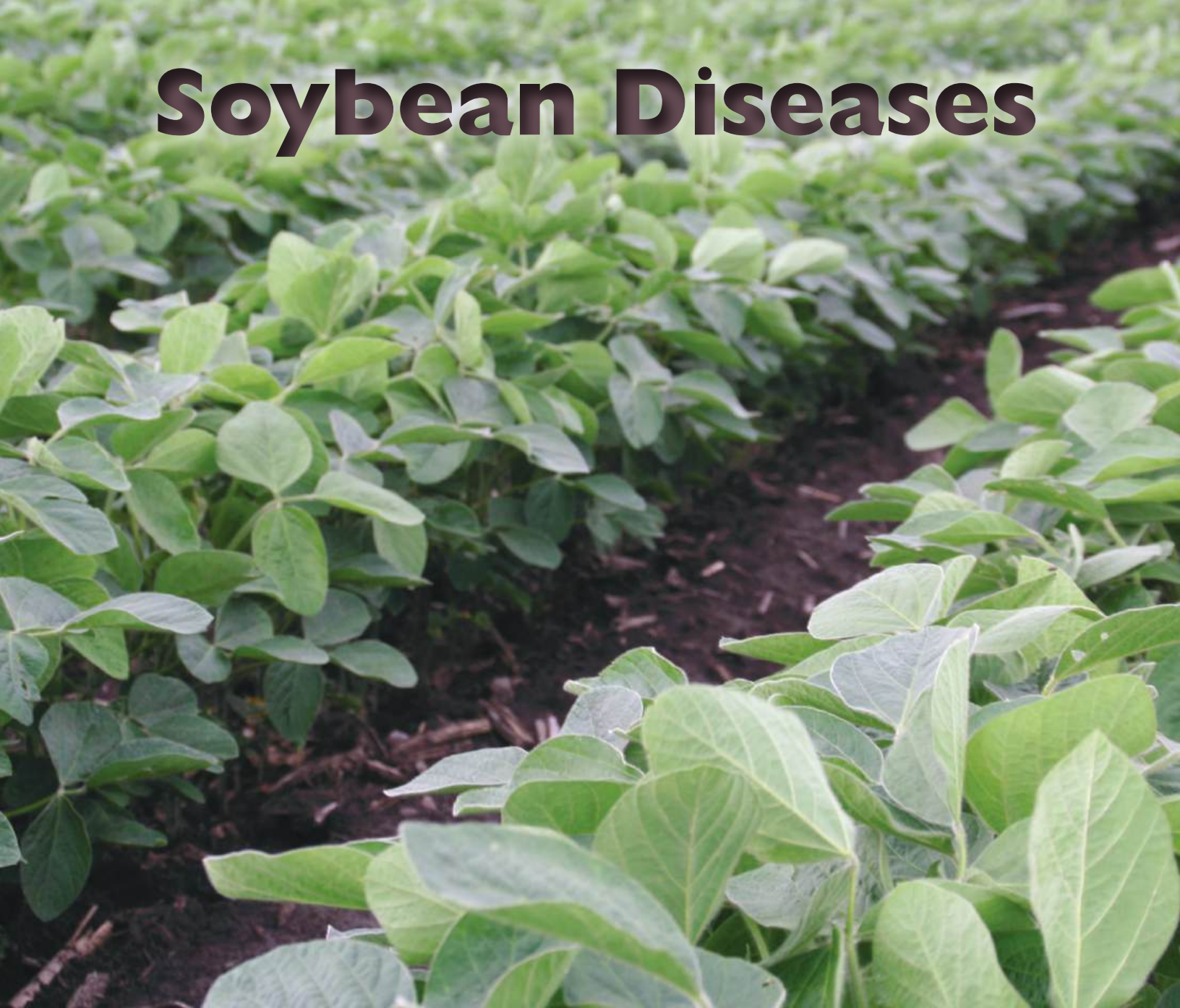


Soybean Diseases



IOWA STATE UNIVERSITY
Extension and Outreach

IPM 0004 January 2018

Soybean Diseases

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Information in this guide may be specific to Iowa. Those using the guide from outside Iowa should check with their state extension service for local recommendations.

Please help us to measure and how this publication has impacted you by completing a short survey at <http://tinyurl.com/soybean-diseases> or by scanning the QR code.



Diseases reduce soybean yields every year to some extent and are also responsible for reductions in grain and seed quality. Disease losses vary from year to year, and their occurrence is strongly influenced by weather conditions.



The purpose of this booklet is to help soybean farmers and other professionals in the agriculture industry identify common diseases and provide general recommendations for disease management. The best disease management strategy can vary depending on each individual situation, so it is wise to seek additional information when you have a disease problem. This publication includes diseases that occur on a regular basis in the Midwest and those that occur less regularly but are potentially very damaging.



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Foliar Diseases

There are several soybean foliar diseases that are likely to appear every year in Iowa. These diseases will reduce the ability of the plant to intercept sunlight and thus diminish photosynthetic capabilities of the plant.



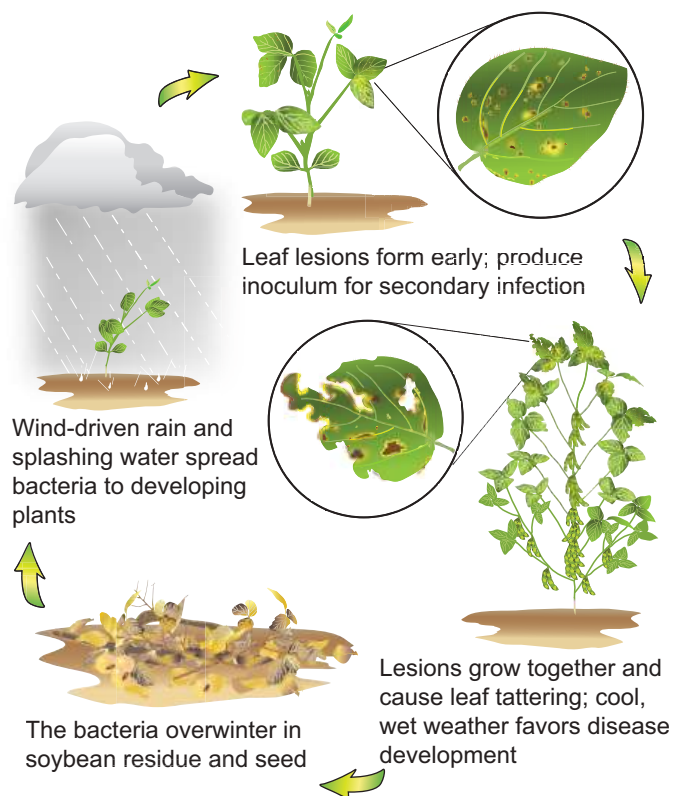
Bacterial Blight (*Pseudomonas savastanoi* pv. *glycinea*)

Along with Septoria brown spot (page 6), bacterial blight is usually one of the first foliar diseases to occur on soybean. Bacterial blight seldom causes serious yield loss. Symptoms usually begin in the upper canopy because young leaves are most susceptible. Small, angular, reddish-brown lesions are surrounded by a yellow halo. As the disease progresses, lesions often grow together to produce large, irregularly shaped dead areas. Centers of older lesions frequently fall out, causing leaves to appear tattered.

The bacteria survive winters in crop residue and seed and are spread by rain and wind. Bacteria can survive on leaf surfaces during the season and infect plants when conditions are suitable. Infection occurs through stomates (natural leaf openings allowing air exchange) and wounds caused by wind, hail, etc. Thus, outbreaks are common after rainstorms with high winds. Cool temperatures favor bacterial blight; warmer temperatures will slow or stop disease development.

Crop rotation and tillage reduce survival of the bacteria. Non-hosts include alfalfa, corn, and small grains; other legume crops can be alternative hosts.

Bacterial Blight Disease Cycle



Best time to scout	V2 through R6
Scouting tip	Scout fields that have received heavy rains
Confused with	Septoria brown spot (p. 6), Bacterial pustule (p. 7), Downy mildew (p. 9), Soybean rust (p. 12), Soybean vein necrosis (p. 16), Insect and mite injury



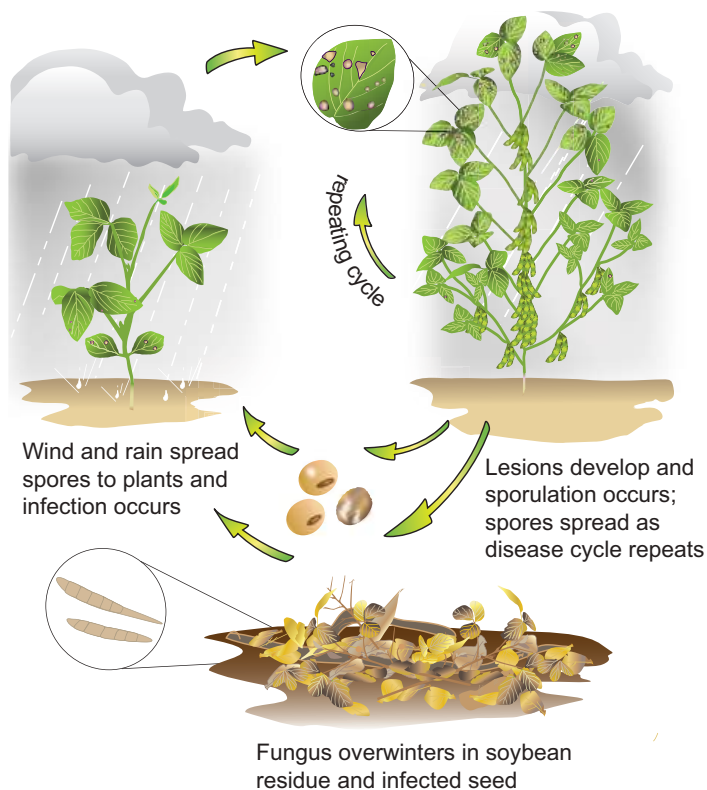
Frog-eye Leaf Spot (*Cercospora sojina*)

Frog-eye leaf spot can cause significant yield loss when widespread within a field. Leaf lesions are small, irregular to circular in shape, and gray with reddish-brown borders. Most commonly occurring on the upper leaf surface, lesions start as dark, water-soaked spots that vary in size. As lesions age, the central area becomes gray to light brown with dark, red-brown margins. In severe cases, disease can cause premature leaf drop and will spread to stems and pods.

The fungus survives in infested crop residue and infected seed. Early season infections contribute to infection of foliage and pods later in the season. Warm, humid weather promotes spore production, infection, and disease development. Young leaves are more susceptible to infection than older leaves, but visible lesions are not seen on young, expanding leaves because the lesions take two weeks to develop after infection.

Resistant soybean varieties are available and should be used where disease is a potential problem. Several races of the pathogen have been identified, and varieties with resistance to all known races are available. Rotating to a non-host crop and tillage will reduce survival of *C. sojina*. Crops not susceptible to this pathogen are alfalfa, corn, and small grains. Foliar fungicides applied during late flowering and early pod set to pod filling stages can reduce frog-eye leaf spot.

Frog-eye Leaf Spot Disease Cycle



Best time to scout R3 through R6; after frequent rains

Scouting tip Check plants in areas where moisture collects from extended dew periods. Found more frequently in continuous soybean fields

Confused with Phyllosticta leaf spot (p. 10), Herbicide injury



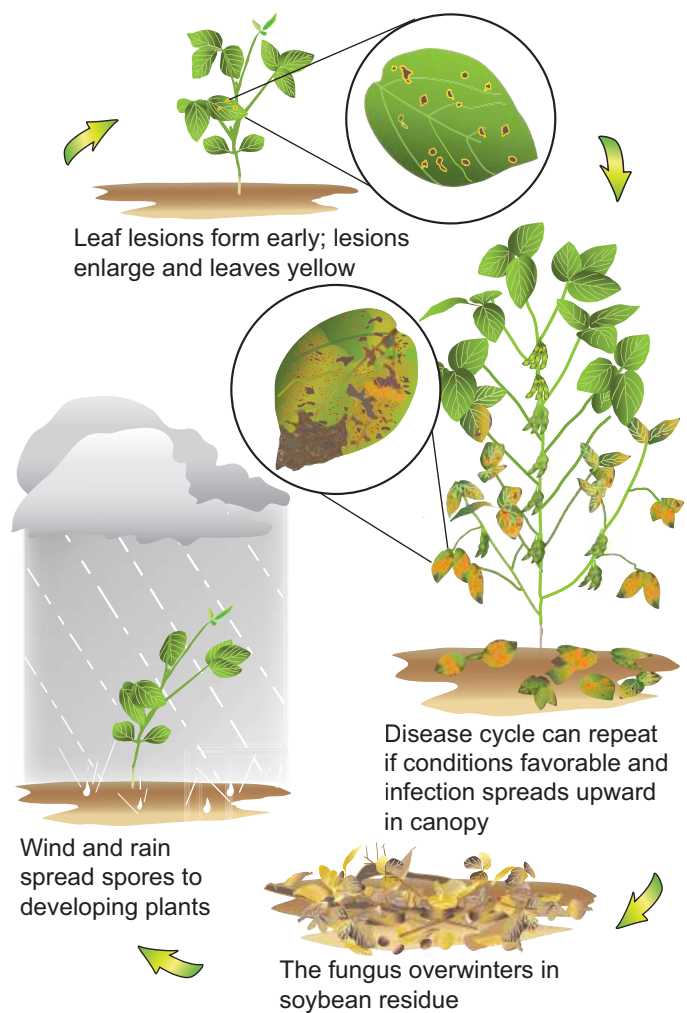
Septoria Brown Spot (*Septoria glycines*)

Septoria brown spot is the most common foliar disease of soybean. Disease develops soon after planting and is usually present throughout the growing season. Symptoms are typically mild during vegetative growth stages of the crop and progress upward from lower leaves during grain fill. Infected young plants have purple lesions on the unifoliate leaves. Lesions on later leaves are small, irregularly shaped, dark brown, and are found on both leaf surfaces. Adjacent lesions can grow together and form larger, irregularly shaped blotches. Infected leaves quickly turn yellow and drop. Disease starts in the lower canopy and, if favorable conditions continue, will progress to the upper canopy. Yield losses depend on how far up the canopy the disease progresses during grain fill.

The fungus survives on infected leaf and stem residue. Warm, wet weather favors disease development. Disease usually stops developing during hot, dry weather but may become active again near maturity or when conditions are more favorable.

There are no known sources of resistance, but differences in susceptibility occur among soybean varieties. The host range of the pathogen includes some other legume species and common weeds such as velvetleaf. Rotation to non-host crops such as alfalfa, corn, and small grains and incorporation of infested crop residue into the soil will reduce survival of *S. glycines*. Foliar fungicides labeled for Septoria brown spot control are available. Applications made during R3 through R5 soybean growth stages may slow the rate of disease development into the middle and upper canopy and protect yield.

Septoria Brown Spot Disease Cycle



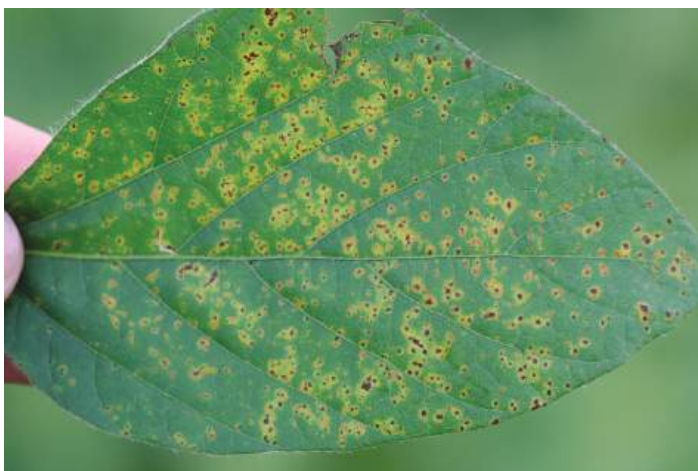
Best time to scout V2 through R6

Scouting tip This disease is found in most fields in Iowa and starts in the lower canopy.

Confused with Bacterial blight (p. 4), Bacterial pustule (p. 7), Downy mildew (p. 9), Phyllosticta leaf spot (p. 10), Soybean rust (p. 12), Soybean vein necrosis (p. 16), Herbicide injury

Occasional or Minor Foliar Diseases

Occasional or minor foliar diseases appear either infrequently (e.g., *Phyllosticta* leaf spot) or do not cause significant yield loss in Iowa in most situations, even though a high incidence of disease may be observed (e.g., *Cercospora* leaf blight). During certain years or with specific field conditions, these diseases may become more problematic.



Bacterial Pustule

(*Xanthomonas axonopodis* pv. *glycines*)

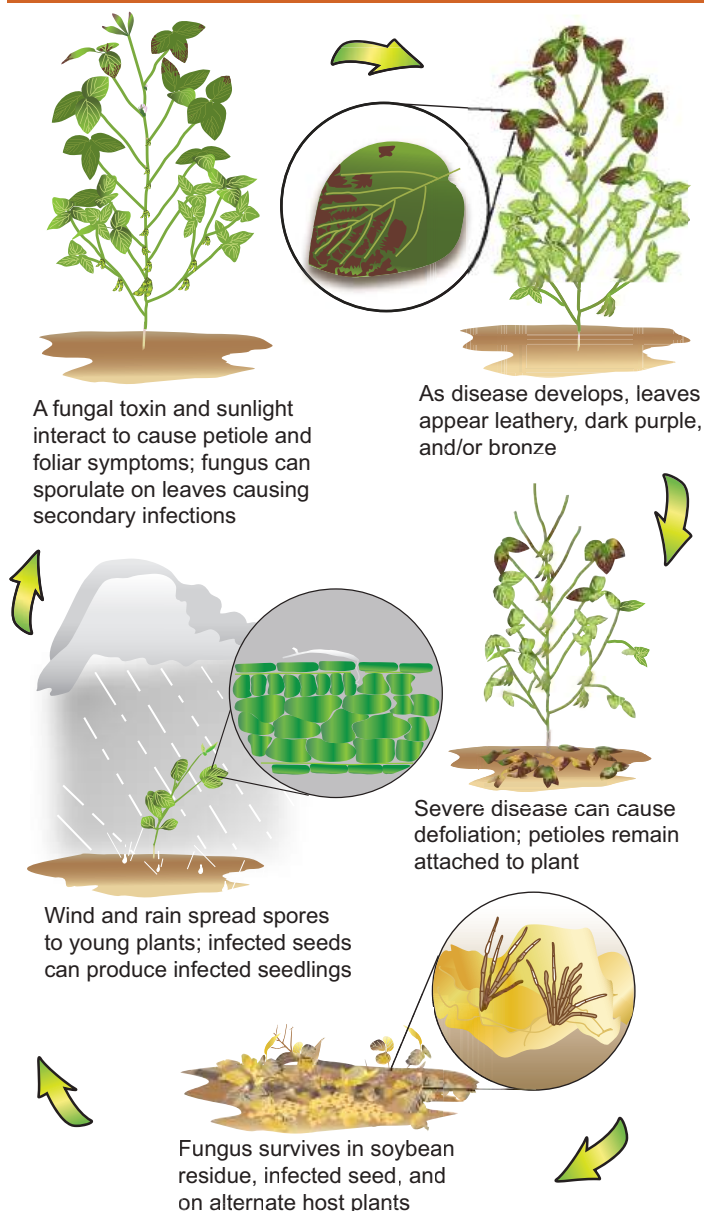
Bacterial pustule occurs mid- to late season when temperatures are warmer. Lesions are found on outer leaves in the mid- to upper canopy. Lesions start as small, pale green specks with elevated centers and develop into large, irregularly shaped infected areas. A greenish-yellow halo surrounds each lesion. A pustule may form in the center of some lesions, usually on the lower leaf surface. Bacterial pustule will not cause leaves to tatter like bacterial blight. Disease development is favored by hot, wet weather. Avoid extremely susceptible varieties where disease is a potential problem; some varieties are marketed as resistant to bacterial pustule. Crop rotation and tillage reduce survival of the bacteria.



Best time to scout	R1 through R6
Scouting tip	Bacterial pustule does not cause leaves to tatter; distinguishing it from bacterial blight. Because of the small blister; this can be confused with soybean rust.
Confused with	Bacterial blight (p. 4), Septoria brown spot (p. 6), Soybean rust (p. 12), Soybean vein necrosis (p. 16)



Cercospora Leaf Blight Disease Cycle

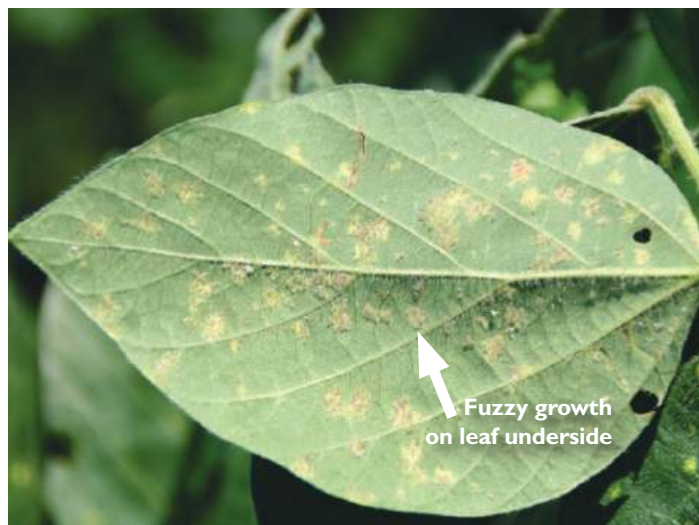


Cercospora Leaf Blight (*Cercospora kikuchii*)

Cercospora leaf blight is frequently seen but rarely causes yield loss. Foliar symptoms usually are seen at the beginning of seed set and occur in the uppermost canopy on leaves exposed to the sun. Leaves are typically only discolored on the upper surface with symptoms ranging from light purple, pinpoint spots to larger, irregularly shaped patches. Affected leaves may become leathery and dark purple with bronze highlights. Symptoms may be confused with sunburn, which typically occurs on the leaf underside. Warm and wet weather favors infection and development of disease. There are commercially available varieties resistant to Cercospora leaf blight. Foliar fungicides are registered for Cercospora leaf blight but are often not needed for this disease.

Although caused by the same organism, there is no consistent relationship between occurrence of Cercospora leaf blight and purple seed stain (page 30).

Best time to scout	R3 through R6
Scouting tip	Check plants in areas of fields where moisture collects from extended dew periods.
Confused with	Soybean rust (p. 12), Soybean vein necrosis (p. 16), Herbicide injury, Sunscald

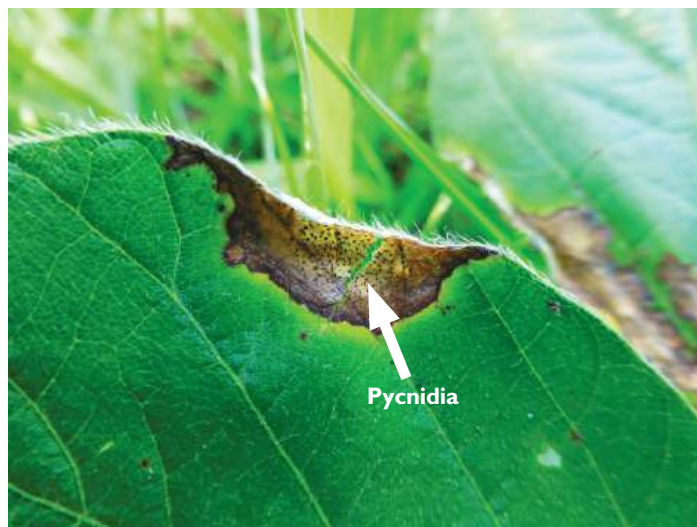


Downy Mildew (*Peronospora manshurica*)

Downy mildew is common but seldom causes serious yield loss. Lesions occur on upper leaf surfaces as irregularly shaped, pale green to light yellow spots that enlarge into pale to bright yellow spots. Older lesions turn brown with yellow-green margins. Young leaves are more susceptible, so disease is often found in the upper canopy. On the leaf underside, fuzzy, gray tufts may be seen growing from lesions, particularly when humidity is high or leaves are wet (e.g., early in the morning). High humidity or extended leaf wetness and moderate temperatures favor disease development. The increased resistance of older leaves and higher temperatures midseason usually stop disease development before extensive damage occurs. Many sources of resistance are available. Crop rotation and burial of infested crop residue can reduce pathogen levels. See page 29 for information about seed symptoms.



Best time to scout	R3 through R6; after frequent rains
Scouting tip	Disease is often found in the upper canopy. Fuzzy growth often can be seen in the morning when dew is still on leaves.
Confused with	Bacterial blight (p. 4), Septoria brown spot (p. 6), Powdery mildew (p. 11), Soybean rust (p. 12), Alfalfa mosaic (p. 13), Soybean vein necrosis (p. 16), Phomopsis seed decay (p. 29)



Phyllosticta Leaf Spot (*Phyllosticta sojaicola*)

Lesions most often occur on leaves and are circular, oval, and irregular or V-shaped. Lesions appear gray or tan and have a narrow, dark margin. In older lesions, numerous small, black specks (i.e., pycnidia) may be visible. The fungus likely survives in infested crop residue and can be seed transmitted. Cool, moist conditions favor disease development. Crop rotation and tillage will reduce survival of the fungus.

Best time to scout	V5 through R6
Scouting tip	A minor disease of soybeans. More common in continuous soybean fields.
Confused with	Frogeye leaf spot (p. 5), Septoria brown spot (p. 6), Soybean vein necrosis (p. 16), Nutrient deficiency, Herbicide injury



Powdery Mildew (*Erysiphe diffusa*)

The most common and characteristic sign of powdery mildew is white, powdery fungal growth that can cover all aboveground plant parts, particularly the upper leaf surface. Later symptoms may include plant tissue yellowing and premature defoliation. Powdery mildew usually does not appear until mid- to late reproductive stages of the crop and can be more severe in late-planted soybeans. The fungus survives in infested crop residue and favorable conditions include cool, cloudy weather and low humidity. Most commercial soybean varieties are resistant. Fungicides effectively manage powdery mildew; however, limited situations exist where fungicide use will be profitable.



Best time to scout	R3 through R6
Scouting tip	This disease is more prevalent in late-planted fields.
Confused with	Downy mildew, (p. 9) Soybean vein necrosis (p. 16)



Soybean rust (*Phakopsora pachyrhizi*)

Soybean rust has not been a major issue for Iowa farmers. However, if conditions are conducive for disease to spread to Iowa during early reproductive stages, significant yield loss may occur. Symptoms are most common after flowering, beginning on lower leaves. Lesions start to form on lower leaf surfaces as small, gray spots that change to tan or reddish-brown. Lesions are scattered within yellow areas appearing translucent if held up to the sun. Mature lesions contain one to many small pustules, usually on lower leaf surfaces. The rust pathogen survives in southern states. Movement northward depends on dispersal of spores and disease establishment in new areas, a process that needs to be repeated several times in a growing season for rust to cause an epidemic in Iowa. Cool, wet weather or high humidity favor epidemics. Dense canopies provide ideal conditions for disease development. Foliar fungicides are the only viable option for managing soybean rust and need to be sprayed prior to infection or, at the latest, very soon after initial infection.



Best time to scout	Track movement of the pathogen at http://sbr.ipmpipe.org to know when to begin scouting.
Scouting tip	Check lower canopy leaves and plants in areas with distinct leaf yellowing or dense canopy.
Confused with	Bacterial blight (p. 4), Septoria brown spot (p. 6), Bacterial pustule (p. 7), Cercospora leaf blight (p. 8), Downy mildew (p. 9)

Viruses



Alfalfa Mosaic
(*Alfalfa mosaic virus*)

Symptoms caused by *Alfalfa mosaic virus* (AMV) are patterns of bright yellow and dark green leaf tissue. Newly emerged leaves may be small with bright yellow spots and brown discoloration and plants may be stunted. Alfalfa mosaic is transmitted by more than 15 aphid species, including the soybean aphid. Alternate hosts of AMV include alfalfa, other legumes, and solanaceous (nightshade family) plants. Alfalfa mosaic may be more prevalent at field edges, especially bordering alfalfa. Current soybean varieties differ in tolerance to alfalfa mosaic based on the degree of symptom expression. Foliar insecticides are likely not effective in reducing transmission of AMV by aphids.



Best time to scout	Any time during the growing season
Scouting tip	Check plants on the edge of fields and also those near alfalfa.
Confused with	Downy mildew (p. 9), Nutrient deficiency, Herbicide injury, Insect and mite injury



Bean Pod Mottle (*Bean pod mottle virus*) and Soybean Mosaic (*Soybean mosaic virus*)

Foliar symptoms caused by *Bean pod mottle virus* (BPMV) (top two images) and *Soybean mosaic virus* (SMV) (right two images) include distorted and wrinkled leaves that have a mottled color pattern. Symptoms appear more severe on young leaves, sometimes with a raised, blistered, or distorted appearance. Symptoms are most obvious at cooler temperatures and often disappear when it is hot. Infection by BPMV decreases pod formation, reduces seed size, weight and number, and may cause seed mottling. The major concern with soybean mosaic is reduced seed quality due to mottled seeds; yield is generally not affected. See page 30 for information about seed symptoms. BPMV is transmitted primarily by the bean leaf beetle while SMV is transmitted by more than 30 aphid species, including the soybean aphid. However, infected seed is the most important way SMV is introduced into a field.



Soybean varieties differ in BPMV tolerance, although the differences are not clearly studied. Consider planting insecticide-treated seed if overwintering survival of bean leaf beetles is predicted, or if bean pod mottle has been confirmed in fields previously and bean leaf beetles have been present. Foliar-applied insecticides can manage bean leaf beetle populations and may reduce bean pod mottle. Foliar insecticides are not effective in reducing transmission of SMV by aphids, even though aphid populations may be reduced.

Best time to scout	Anytime during the growing season; however, the most practical way to check for virus infection is to look at the seed.
Scouting tip	Check plants on the edge of fields. Early-planted fields may be at higher risk for BPMV infection whereas SMV risk is greater for later planted fields.
Confused with	Tobacco streak (p. 17), Nutrient deficiency, Herbicide injury (growth regulator herbicide drift), Insect and mite injury



Soybean Dwarf (Soybean dwarf virus)

Symptoms caused by *Soybean dwarf virus* (SbDV) can vary due to virus strain, infection timing, and variety of soybean. Leaves can be dark green, brittle, wrinkled, curled, small, and thick while green stem, delayed maturity, plant stunting, yellowing between leaf veins, and decreased seed number can also occur. SbDV is transmitted by aphids and has multiple legume and non-legume host plants. Severity and incidence of soybean dwarf can increase if early season environmental conditions favor aphid development. Soybean dwarf-tolerant or -resistant soybean varieties exist. Virus dissemination within a field can be reduced by managing aphids using seed and foliar insecticides. Soybean fields planted next to clover are at higher risk of exposure to SbDV inoculum, while soybean seeds are not a source of inoculum.



Best time to scout	R3 through R6
Scouting tip	Conditions favorable for aphids early in the season favor soybean dwarf.
Confused with	Nutrient deficiency, Insect and mite injury



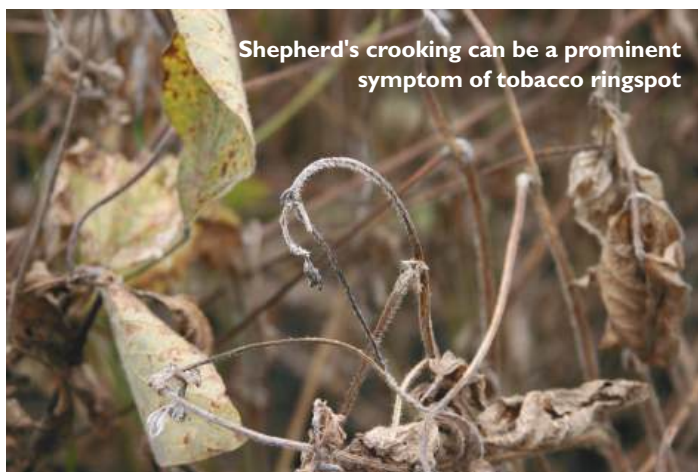
Soybean Vein Necrosis (*Soybean vein necrosis virus*)

Symptoms caused by *Soybean vein necrosis virus* (SVNV) often begin as yellowing near leaf veins that develops into reddish-brown lesions, causing death of leaf tissue. Lesions follow or spread from leaf veins. Leaf veins may limit lesions, resulting in an angular lesion appearance. Clear, yellow, or dark brown discoloration of leaf veins can occur. Browning of veins may be especially noticeable on the leaf underside. SVNV is vectored by the soybean thrips and possibly other thrips, and hot, dry weather that favors thrips may cause increased disease spread. Weeds may provide an overwintering host for SVNV; thrips migrating from the south may also carry the virus.

Best time to scout R1 through R6

Scouting tip Affected plants are generally scattered randomly in a field and distribution of symptoms on a leaf are usually non-uniform. Lesions follow or spread from leaf veins, which can help differentiate soybean vein necrosis from diseases that appear similar.

Confused with Bacterial blight (p. 4), Septoria brown spot (p. 6), Bacterial pustule (p. 7), Cercospora leaf blight (p. 8), Downy mildew (p. 9), Phyllosticta leaf spot (p. 10), Powdery mildew (p. 11), Sudden death syndrome (p. 24), Nutrient deficiency, Herbicide injury, Insect and mite injury



Tobacco Ringspot (*Tobacco ringspot virus*) and Tobacco Streak (*Tobacco streak virus*)

Many symptoms of *Tobacco ringspot virus* (TRSV) (above) and *Tobacco streak virus* (TSV) (right) infection are similar: These symptoms include bud blight, shepherd's crooking, stunting, wilting, delayed maturity, bud and leaf proliferation, reduced pod number; and discolored, distorted, and dead leaves. TRSV-infected plants may have discolored pith, petiole distortion and discoloration, leaf vein discoloration, and poor root nodulation.

TSV infection can cause discoloration and dead streaks at nodes, dark spots on pods, and delayed seed development. TSV and TRSV can be seed transmitted but are also spread by several insects including thrips. Hot, dry weather is conducive to disease development. Make sure to plant disease-free, high-quality seed. Tobacco streak incidence may be decreased by delaying planting while tobacco ringspot may be reduced by planting next to corn fields rather than pastures.

Best time to scout R4 through R7

Scouting tip Affected plants can occur in patches or be scattered randomly in a field. Plants can have green soybean syndrome. Infected plants may not be noticeable until after non-infected plants begin to mature.

Confused with Bean pod mottle (p. 14), Anthracnose stem blight (p. 26), Nutrient deficiency, Herbicide injury, Insect and mite damage

Root and Stem Diseases

There are many diseases of soybean that affect the roots and stems. Some of these diseases also will cause foliar symptoms such as sudden death syndrome.



Diaporthe Complex: Northern Stem Canker (*Diaporthe caulivora*) and

Pod and Stem Blight (*D. sojae*)

Northern stem canker and pod and stem blight are diseases that are part of the *Diaporthe* complex. Stems, petioles, pods, and seeds are all susceptible to infection. For stem canker, initial symptoms are small, reddish-brown lesions that appear soon after flowering near the nodes in the lower third of the soybean canopy. Lesions expand lengthwise and form sunken cankers that are gray-brown with reddish margins. Mature plants can exhibit irregular, thin black lines, or “zone” lines, on the interior of lower stems. For pod and stem blight, the most characteristic sign is linear rows of black specks on mature soybean stems. The

specks are pycnidia and can be seen during the season on prematurely killed petioles or stems.

The fungal pathogens survive in infested residue, seed, or in soil. Certain weeds may serve as hosts to some pathogenic *Diaporthe* species. Infection occurs when spores are splashed by rain onto plants in early vegetative growth stages. Northern stem canker is favored by cooler temperatures and extended periods of rainy weather occurring early in the growing season. Pod and stem blight is favored by warm, humid weather when soybean plants are maturing. Also, disease is more severe if harvest is delayed.

Varieties with resistance to northern stem canker are available but not for pod and stem blight. Crop rotation and tillage will reduce survival of *Diaporthe* species.

Best time to scout	R3 through R6 (northern stem canker); R6 through R8 (pod and stem blight)
Scouting tip	Northern stem canker may be more problematic in no-till fields and fields rotated with alfalfa, which is an alternate host. It is important to scout seed production fields for pod and stem blight as the fungus can be transmitted by seed.
Confused with	Phytophthora root and stem rot (p. 20), Sudden death syndrome (p. 24), White mold (p. 25), Anthracnose stem blight (p. 26), Brown stem rot (p. 27), Charcoal rot (p. 28), Herbicide injury, Fungicide injury, Late season frost damage, Lightning damage



Fusarium Root Rot and Wilt (*Fusarium* spp.)

Fusarium is a very common soil fungus, and more than 10 different species are known to cause root rot. However, the economic impact on yield is not well documented. Infected seedlings exhibit poor or slow emergence and are often stunted. Root rot appears as reddish-brown to dark brown discolored roots and poor nodulation. Foliar symptoms of *Fusarium* wilt include scorching of the upper leaves, while middle and lower canopy leaves can turn chlorotic and later wither and drop from the plant. Infected plants have brown vascular tissue in the roots and stems and show wilting of the stem tips. External decay or stem lesions are not seen above the soil line. Cool, wet soil favors early season infection, while hot, dry conditions favor later season symptoms.

The causal fungi survive in the soil or in plant residue. Certain weeds may serve as hosts to some pathogenic *Fusarium* species. Stresses such as herbicide injury, high soil pH, iron chlorosis, nematode feeding, and nutritional disorders can all predispose plants to infection. After infection, damage to plants can be worsened if soil moisture is limited because of the damaged root systems.

Reducing or eliminating stress factors, such as use of herbicides that cause injury to soybeans, wet soils, and soybean cyst nematode, can help reduce root rot problems. Growing varieties tolerant to iron deficiency chlorosis should be considered if the root rot seems associated with iron deficiency chlorosis. No resistant varieties are available, but fungicide seed treatments may protect seedlings.

Best time to scout	VE through R6; after frequent rains
Scouting tip	Look closely at roots as they often are missing many lateral roots.
Confused with	Phytophthora root and stem rot (p. 20), Pythium seedling blight and root rot (p. 21), Rhizoctonia seedling blight and root rot (p. 22), Herbicide injury, Early season frost damage, Insect and mite injury



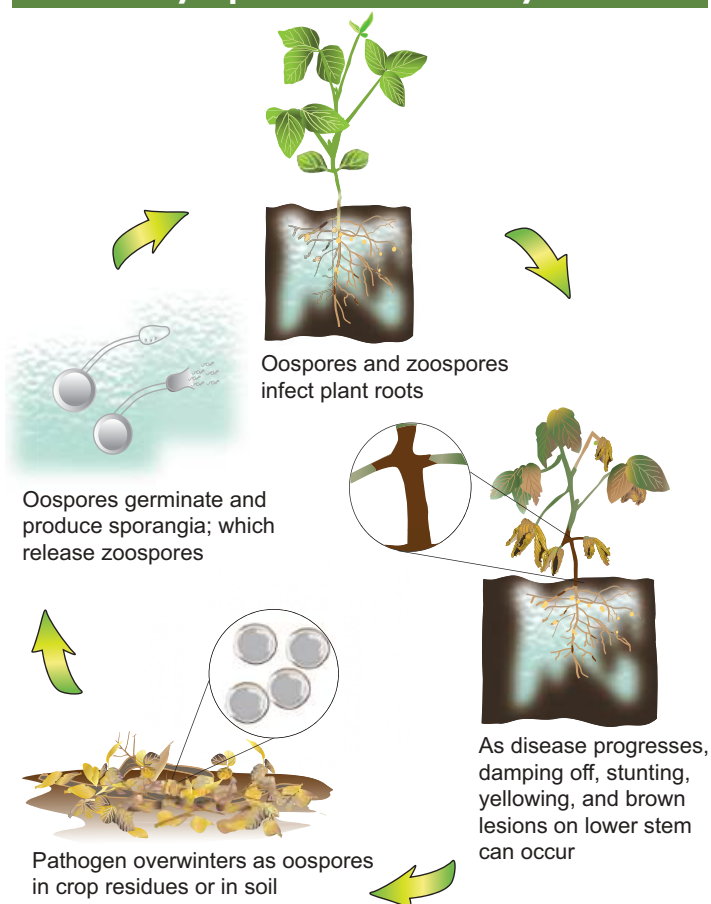
Phytophthora Root and Stem Rot (*Phytophthora sojae*)

Phytophthora sojae can infect soybeans at any growth stage from seed to maturity. Early season symptoms include seed rot and pre- and postemergence damping off. The most characteristic symptom of *Phytophthora* root rot is a dark brown lesion on the lower stem that extends up from the taproot of the plant. The lesion often reaches as high as several nodes and will girdle soybean stems, restricting flow of nutrients and water, and stunting or killing the plant. *Phytophthora* root rot is more severe in no-till fields, poorly drained soils, or low-lying areas that are prone to flooding.

The oomycete pathogen survives on crop residue or in the soil as oospores. When soil temperatures reach 60°F and soil is saturated, oospores germinate and produce swimming spores, called zoospores. Warm, saturated soil, especially shortly after planting, are conducive to disease. Infection occurs via the roots, and from there the pathogen colonizes the roots and stems.

Phytophthora root rot is best managed by planting resistant varieties. There are soybean varieties available that have complete resistance to specific pathotypes (races) of *P. sojae*. There are numerous pathotypes, and many pathotypes can exist in a single field. Partial resistance (resistance made up of more than one gene) is available and is effective against all pathotypes. Where *Phytophthora* root and stem rot is a problem, seed treatments targeting oomycetes can provide some protection.

Phytophthora Disease Cycle



Best time to scout	VE through R6; 1–2 weeks after excessive rains
Scouting tip	Occurs in wet, waterlogged soils and compacted soils
Confused with	Diaporthe complex (stem canker) (p. 18), Fusarium root rot and wilt (p. 19), Pythium seedling blight and root rot (p. 21), Rhizoctonia seedling blight and root rot (p. 22), Sudden death syndrome (p. 24), White mold (p. 25), Anthracnose stem blight (p. 26), Brown stem rot (p. 27), Charcoal rot (p. 28), Herbicide injury, Early season frost damage, Insect and mite injury



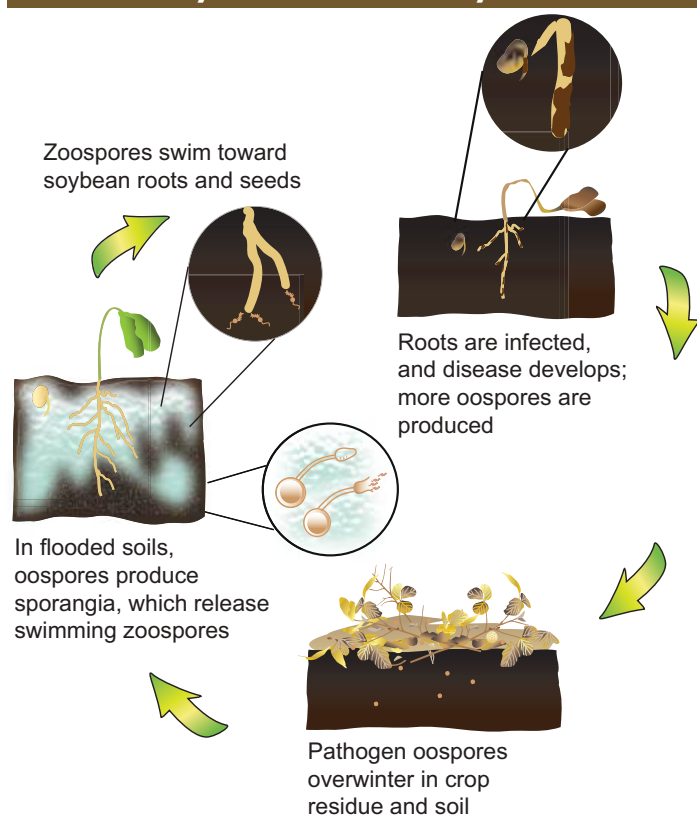
Pythium Seedling Blight and Root Rot (*Pythium* spp.)

Pythium species cause pre- or postemergence damping off. Infected seeds appear rotted and soil sticks to them. Infected seedlings have water-soaked lesions on the hypocotyl or cotyledons that develop into a brown soft rot. Diseased plants are easily pulled from the soil because of rotted roots.

The oomycete pathogens survive as oospores either in plant residue or in soil. Saturated soil is critical for infection for all *Pythium* species. In general, *Pythium* species that are prevalent in the northern United States infect plants at lower temperatures (50 to 60°F), and *Pythium* species in the South infect plants at warmer temperatures (85 to 95°F), although there are exceptions. Early planting dates increase the risk of disease in Iowa. Diseased plants are more common in no-till fields or low-lying areas of the field that are prone to flooding.

Planting in cold, wet soils should be avoided to reduce infection by *Pythium* species that infect at low temperatures. Where *Pythium* is a problem, seed treatments targeting oomycetes can provide some protection. Resistance to metalaxyl/mefenoxam has been documented.

Pythium Disease Cycle



Best time to scout	VE through V2 in areas where soil temperatures have been cold (<60°F) and soils are very wet
Scouting tip	This disease occurs in low and wet areas of the field. <i>Pythium</i> -like symptoms in late-planted soybeans may actually be caused by <i>Phytophthora</i> .
Confused with	Fusarium root rot and wilt (p. 19), Phytophthora root and stem rot (p. 20), Rhizoctonia seedling blight and root rot (p. 22), Herbicide injury, Early season frost damage, Flooding, Insect and mite injury



Rhizoctonia Seedling Blight and Root Rot (*Rhizoctonia solani*)

Pre- and postemergence damping off can occur. Infected seedlings have reddish-brown lesions on the hypocotyls at the soil line. These lesions are sunken and remain firm and dry. The root rot phase may persist into late vegetative to early reproductive growth stages. Symptoms may disappear if infected plants grow out of the root rot problems, although plants may remain stunted.

The fungus survives on plant residue or in soils as sclerotia. Infection may occur soon after seed is planted. Cool, wet springs followed by hot, dry conditions are conducive to disease. Disease is more severe on light and sandy soils, and can be more common on the slopes of fields. Unfortunately, many strains of *Rhizoctonia* can infect corn, alfalfa, dry bean, and some cereal crops, negating the effectiveness of crop rotation.

Eliminating stress factors, such as use of herbicides that cause injury to soybean roots, can help reduce root rot problems. Fungicide seed treatments are effective against *Rhizoctonia*.

Best time to scout	VE through RI
Scouting tip	Check plants in fields with poor emergence or slowed early growth.
Confused with	Fusarium root rot and wilt (p. 19), Phytophthora root and stem rot (p. 20), Pythium seedling blight and root rot (p. 21), Herbicide injury, Early season frost damage, Insect and mite injury



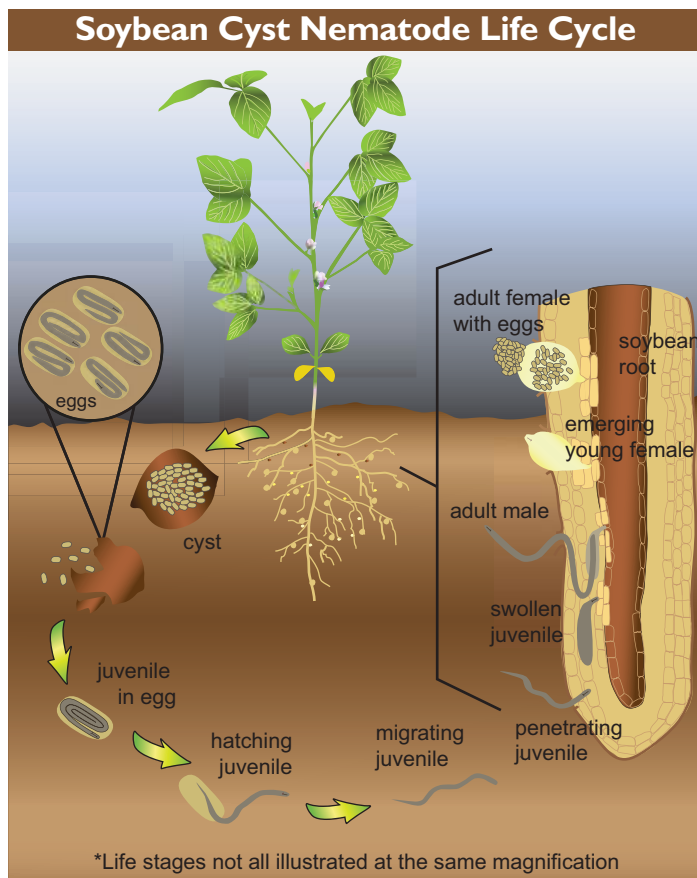
Soybean Cyst Nematode (*Heterodera glycines*)

Damage from the soybean cyst nematode (SCN) may not be obvious in high-yielding fields or during years when soil moisture is plentiful. However, yield losses of 40 percent or more are still possible. When symptoms are associated with damage, symptomatic plants usually occur in patches. Noticeable symptoms of SCN include stunting, slow or no canopy closure, and chlorotic foliage. Infected plants have poorly developed root systems and may have reduced numbers of nitrogen-fixing nodules. Signs of SCN include white females most readily seen in the field starting about six weeks after crop emergence. Dig roots and carefully remove soil to see the females.

SCN survives in the soil as eggs within dead females, called cysts. These eggs can survive several years in the absence of a soybean crop. Conditions that favor soybean growth are also favorable for SCN development. High soil pH (7.0 to 8.0) may be used to predict where SCN is more problematic.

Resistant varieties are available. Resistant varieties are not resistant to all SCN populations and most resistant varieties contain one common source of genetic resistance. Rotating sources of resistance may help prevent development of more damaging SCN populations. Cropping sequences should include non-host crops (usually corn) and resistant soybean varieties. Cleaning soil from tillage and harvest equipment can help keep SCN from spreading to new fields. No-till practices may slow SCN move-

ment and lower population densities. Nematode-protectant seed treatments may provide early season protection. Although proper management can greatly reduce SCN numbers, it is impossible to eliminate SCN from a field.



Best time to scout	Six weeks after planting until three to four weeks before harvest
Scouting tip	Areas where SCN may first appear in a field include near field entrances, areas that have been flooded, weedy areas, high pH spots, and lower yielding areas.
Confused with	Sudden death syndrome (p. 24), Charcoal rot (p. 28), Nutrient deficiency, Drought, Flooding



Petioles of infected plants remain on stem after leaves fall

Sudden Death Syndrome (*Fusarium virguliforme*)

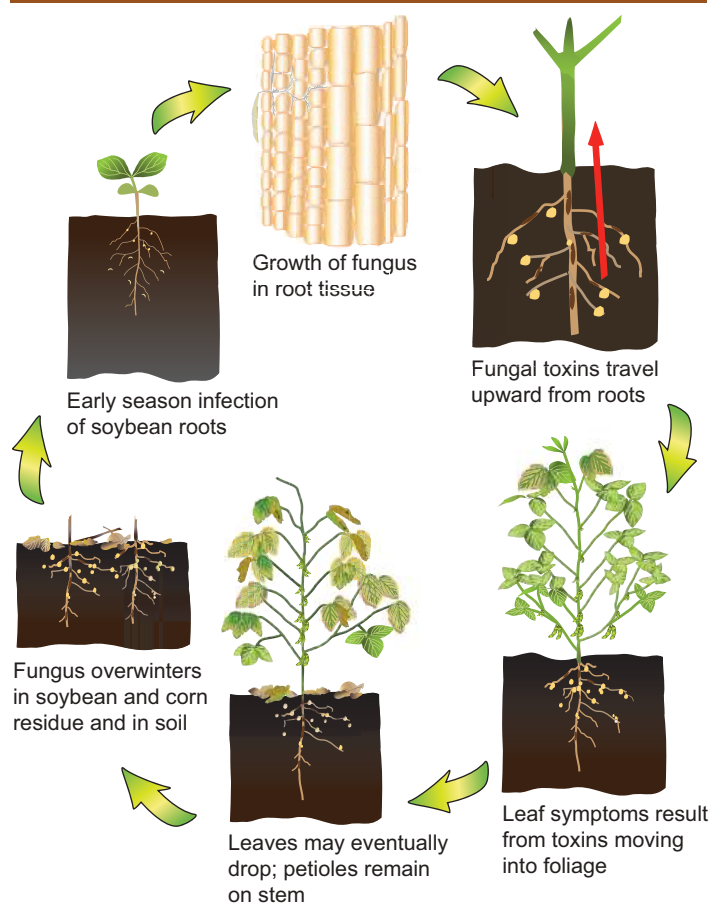
Foliar symptoms of sudden death syndrome (SDS) are a result of a toxin, produced by the fungus, moving from roots to the leaves. Foliar symptoms rarely appear until after flowering. Leaves of infected plants initially show scattered yellow spots between leaf veins. Spots grow to form large chlorotic and necrotic blotches between the leaf veins, while the midvein and major lateral veins remain green. Leaflets eventually drop, but petioles remain on the stem. Diseased plants have rotted taproots and lateral roots. When stems are cut lengthwise, the woody tissue of the taproot is discolored light gray to brown. This discoloration may extend up to two inches above ground. Bluish fungal growth may be seen on the surface of roots if soil moisture is high. Foliar symptoms of SDS look similar to those of brown stem rot (BSR). A good way to distinguish between BSR and SDS is the presence or absence of internal stem browning. Stems of SDS-infected plants have white pith, while BSR causes brown discoloration of the pith.

The fungus survives on infested crop residue or in soil for several years. Crop rotation with corn is not effective because the fungus can increase on corn residue. Root infection in soybean can occur within days of planting and is favored by high soil moisture. Significant rainfall during reproductive stages favors foliar symptom development. Soil compaction and infection by soybean cyst nematode (SCN) can increase disease severity.

Several soybean varieties have partial resistance (resistance made up of more than one gene) to SDS. Check with a local seed dealer to identify an appropriate variety. Effective seed

treatments also can help reduce SDS. Reducing soil compaction and SCN population densities can decrease the risk of SDS.

Sudden Death Syndrome Disease Cycle



Best time to scout	R1 through R6
Scouting tip	Check plants in areas with high soil moisture and in fields infested with soybean cyst nematode. Split stems to distinguish from brown stem rot.
Confused with	Soybean vein necrosis (p. 16), Diaporthe complex (stem canker) (p. 18), Phytophthora root and stem rot (p. 20), Soybean cyst nematode (p. 23), White mold (p. 25), Brown stem rot (p. 27), Charcoal rot (p. 28), Nutrient deficiency, Herbicide injury, Fungicide injury



White Mold (Sclerotinia Stem Rot) (*Sclerotinia sclerotiorum*)

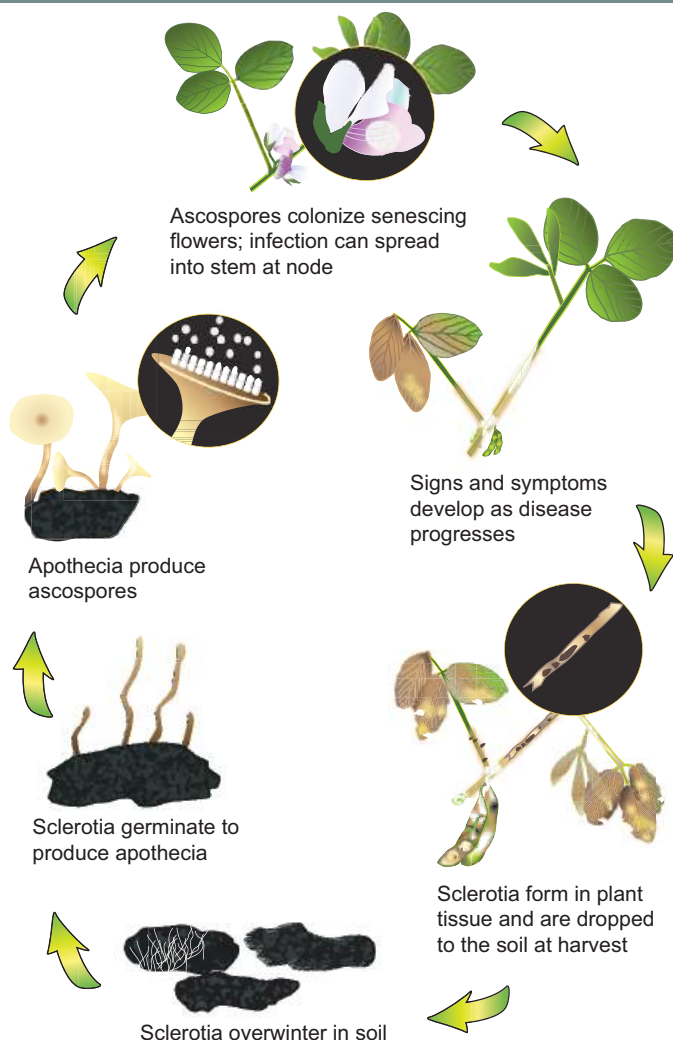
White mold is often recognized by fluffy, white growth on soybean stems. Initial symptoms generally develop from R3 to R6 as gray to white lesions at the nodes. Lesions rapidly progress above and below the nodes, sometimes girdling the stem. White, fluffy mold soon covers the infected area, especially during periods of high relative humidity. Characteristic black sclerotia eventually are visible and embedded within mold on stem lesions, and inside the stem as the plant approaches death. Initial foliar symptoms include leaf tissues between major veins turning a gray-green cast; eventually leaves die and turn completely brown while remaining attached to the stem. See page 30 for information about seed symptoms.

The fungus survives in the soil for several years as sclerotia. The disease cycle begins when mushroom-like structures called apothecia are formed on the soil surface from sclerotia. Spores from apothecia infect senescing soybean flowers and the fungus eventually grows to the stem. The disease is more prevalent during cool, wet, or humid seasons and in fields where the canopy closes during soybean flowering and early pod development.

No soybean variety is completely resistant to white mold, but varieties do range from moderately resistant to very susceptible. At least two or three years of a non-host crop can reduce the number of sclerotia in soil. Crops that should not be in rotation with soybean in fields with white mold risk are beans, peas, sunflowers, and cole crops (mustard or Brassicaceae family).

More sclerotia are found near the soil surface in no-till systems, but sclerotia numbers begin to decline if left undisturbed. Early planting, narrow row width, and high plant populations all accelerate canopy closure and favor disease development. However, modification of these practices also may reduce yield potential. Weed control is critical as many broadleaf weeds are hosts of the white mold pathogen. Also, some herbicides may suppress the activity of the fungus or disrupt germination of sclerotia. Some antagonistic fungi may be applied to the soil to colonize and reduce sclerotia numbers. Foliar fungicides can manage white mold, or at least reduce disease severity; however, application timing is critical. Fungicides are most effective when applied immediately before infection.

White Mold Disease Cycle



Best time to scout	R4 through R6
Scouting tip	Check plants in areas where moisture collects due to fog and extended dew periods and also in pockets of low air movement.
Confused with	Diaporthe complex (stem canker) (p. 18), Phytophthora root and stem rot (p. 20), Sudden death syndrome (p. 24), Brown stem rot (p. 27), Charcoal rot (p. 28), Late season frost damage

Occasional or Minor Root and Stem Diseases

Several other soybean root and stem diseases are also found in Iowa. However, yield loss from these diseases is generally minimal as they occur either sporadically (e.g., brown stem rot) or usually do not cause significant losses despite a high incidence (e.g., anthracnose stem blight).



Anthracnose Stem Blight
(*Colletotrichum truncatum*)

Anthracnose stem blight is generally a late season disease prevalent on maturing soybeans. Irregularly shaped red to dark brown blotches on stems and petioles can appear during early reproductive stages. Petiole infection can result in a shepherd's crook. Near maturity, black fungal bodies that produce small, black spines and spores are evident on infected stems, petioles, and pods. Warm, wet weather favors infection and disease development.

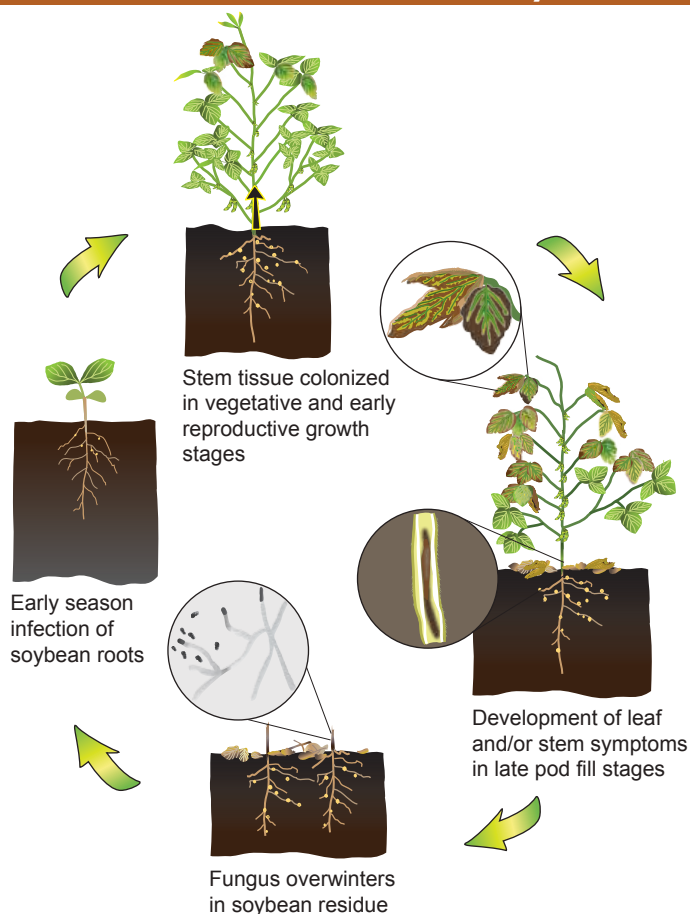
Crop rotation and tillage can reduce inoculum. Foliar fungicides made during the early to mid-reproductive growth stages can reduce disease, although there are limited situations where fungicide use targeting anthracnose stem blight will be profitable.



Best time to scout	R5 through R7
Scouting tip	It is especially important to scout seed production fields because the fungus that causes the disease can be transmitted by seed.
Confused with	Tobacco ringspot (p. 17), Diaporthe complex (pod and stem blight and stem canker) (p. 18), Phytophthora root and stem rot (p. 20), Charcoal rot (p. 28), Late season frost damage



Brown Stem Rot Disease Cycle



Brown Stem Rot (*Cadophora gregata*)

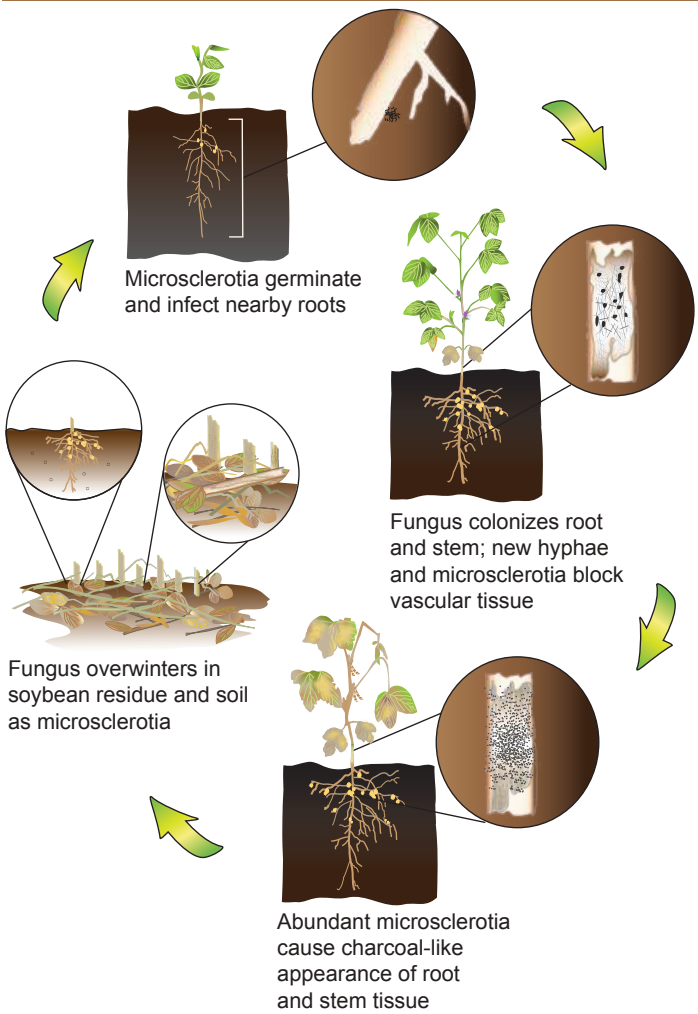
Characteristic foliar symptoms of brown stem rot (BSR) include chlorosis and necrosis between leaf veins, followed by leaf curling and leaf death. Leaf symptoms vary depending on soybean variety, fungal strain, and environmental conditions. In some instances, no foliar symptoms occur. Foliar symptoms can be similar to those of sudden death syndrome and stem canker and appear after early pod set. Stem symptoms usually occur prior to leaf symptoms and can occur even if foliar symptoms never appear. Externally, infected stems look healthy. However, when stems are split lengthwise, internal browning of vascular tissue and pith is evident, especially at nodes and in the lower stem. BSR is more severe when temperatures are cool and adequate soil moisture is present. Foliar symptoms are particularly sensitive to environmental conditions during reproductive growth stages of the crop and are suppressed when temperatures are high or soil is dry during these stages.

The fungus does not produce long-term survival structures but can reproduce on soybean residue throughout the winter. The pathogen has a limited host range, so crop rotation to a non-host crop (corn, small grains, and forage legumes) will reduce pathogen levels. The most efficient way to manage BSR is using resistant varieties. Soybean cyst nematode (SCN) breaks resistance to BSR in most BSR-resistant varieties. However, many soybean varieties with PI 88788-derived SCN resistance also have BSR resistance. Varieties with Peking- or Hartwig-derived SCN resistance, however, may be susceptible to BSR.

Best time to scout	R4 through R6
Scouting tip	Severity is greater in soils with low phosphorous and potassium and a soil pH below 6.5. Often occurs with soybean cyst nematode, which may increase severity of brown stem rot.
Confused with	Diaporthe complex (stem canker) (p. 18), Phytophthora root and stem rot (p. 20), Sudden death syndrome (p. 24), White mold (p. 25), Charcoal rot (p. 28), Herbicide injury, Fungicide injury



Charcoal Rot Disease Cycle



Charcoal Rot (*Macrophomina phaseolina*)

Initial symptoms of charcoal rot are patches of stunted or wilted plants. Leaves remain attached after plant death. The fungus produces numerous tiny, black fungal structures called microsclerotia that are scattered throughout the pith and on the surface of taproots and lower stems. These microsclerotia give the lower stem and taproots a discolored light gray or charcoal-like appearance. Although infection can occur very early in the season, symptoms usually appear after flowering. Charcoal rot is most yield-limiting when weather conditions are hot and dry.

The fungus survives in soil or soybean residue as microsclerotia. Growing small grains, such as wheat or barley, can reduce microsclerotia numbers. Because corn is a host, it will not reduce levels of the fungus when grown in rotation with soybeans. Fields with minimal or no tillage may have fewer symptoms because of lower soil temperatures and greater water-holding capacity. Avoid excessive seeding rates so that plants do not compete for moisture, which increases disease risk during a dry season.

Best time to scout	R6 through R8
Scouting tip	The driest areas of the field will show symptoms first.
Confused with	Diaporthe complex (pod and stem blight and stem canker) (p. 18), Phytophthora root and stem rot (p. 20), Soybean cyst nematode (p. 23), Sudden death syndrome (p. 24), White mold (p. 25), Anthracnose stem blight (p. 26), Brown stem rot (p. 27), Late season frost damage, Drought, Lightning damage

Seed Diseases

Seed diseases of soybean can result in a variety of problems, including lowering the value of marketed grain. If infected seed is planted, poor emergence and spread of disease can occur. Several soybean diseases that cause foliar symptoms also cause seed symptoms; disease symptoms may not be present even if seed is infected.



Downy Mildew (*Peronospora manshurica*)

Downy mildew-infected pods show no external symptoms, but the inside of the pod and seed may be covered with a dried, whitish fungal mass that appears crusty and contains spores (above). Infected seed can be smaller, appear dull white, and have cracks in the seed coat.



Phomopsis Seed Decay (*Diaporthe longicola*)

Phomopsis-infected seeds are cracked and shriveled and are often covered with chalky, white mold. If infected seeds are planted, emergence may be low due to seed rot or seedling blight. Infected seedlings have reddish-brown, pinpoint lesions on the cotyledons or reddish-brown streaks on the stem near the soil line.

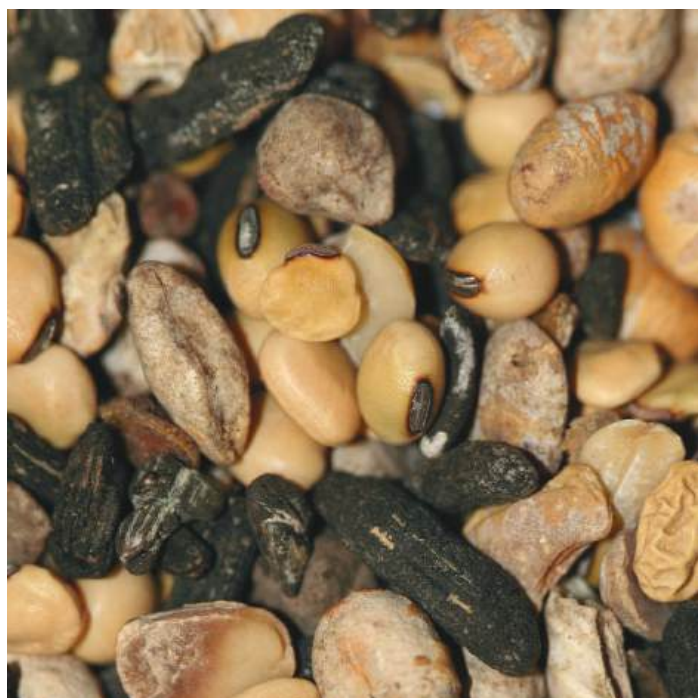


Purple Seed Stain (*Cercospora kikuchii*)

Seeds infected with the pathogen that causes purple seed stain can be symptomless or have pink to purple spots extending from the hilum (above). Crop value may be lowered by dockage or seed certification denial, but yield is not reduced.

Viruses (*Bean Pod Mottle Virus*, *Soybean Mosaic Virus*)

Virus infection can cause diffusion of pigments from the seed hilum, resulting in mottling of the seed coat. Hilum color determines the color and intensity of seed coat discoloration (above). Mottling is associated with poor germination and may result in a grain grade reduction, particularly for food grade soybeans.

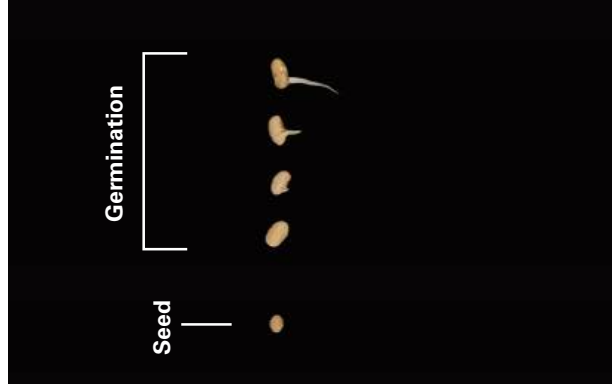


White Mold (*Sclerotinia sclerotiorum*)

Pods affected by white mold generally contain seeds that are smaller, lighter, white, and cottony (above). Seeds may be replaced by sclerotia. It is unusual for an infected seed to look normal. Soybean seed lots can be contaminated with sclerotia, the survival structures produced by *S. sclerotiorum* in infected plants (right).

SOYBEAN GROWTH and DEVELOPMENT

Soybean has an indeterminate growth habit, meaning reproductive stages overlap with vegetative growth and other reproductive stages. Because of this, soybean has the ability to compensate for stand losses and adjust to stress conditions from competition for sunlight, water, and nutrients. This makes soybean less sensitive to crop yield loss than other crops such as corn.



- ▶ Seed imbibes ~50% of its weight in water.
- ▶ Sensitive to imbibitional chilling within 12 – 36 hours of planting.
- ▶ Soil crusting can result in breakage of the hypocotyl.



- ▶ Upon emergence, plants are immediately sensitive to hail damage and frost (temperature below 29°F).
- ▶ Stems severed below the cotyledons result in direct stand loss; stems severed above nodes will result in branch growth from axillary buds.



- ▶ The plant is no longer relying on the cotyledon nutrient/food reserves.
- ▶ Nodule production is initiated.

SOYBEAN GROWTH and DEVELOPMENT (continued)

5th and 6th trifoliate stage
(V5 – V6)



- ▶ Roots typically cross 30-inch inter-row spacing.
- ▶ Rapid growth and nutrient uptake begin at this stage.

Flowering and pod set
(R1 – R4)



- ▶ Flowering is expected to begin between June 20 and July 5; 60 to 75% of flowers will abort.
- ▶ Greater grain yields are typically achieved when more vegetative biomass is produced before flowering begins.
- ▶ Plant stresses from heat, moisture, pesticides, diseases, and insects as well as limited carbohydrate supply from photosynthesis can cause flower and pod abortion.

Seed filling
(R5 – R6)



- ▶ High amounts of sunlight, water, and nutrients are critical for seed fill (seeds per pod and seed size), yield components, and determination of seed oil and protein content.
- ▶ Hail injury during this timeframe can cause grain quality issues.
- ▶ At this point, dry matter is allocated to the seeds; defoliation events in this timeframe can reduce yields by up to 75%.

Maturity
(R7 – R8)



- ▶ Stress and stage length have a minimal effect on grain yield because seeds per pod and seed size are largely determined already.
- ▶ Hail events can greatly reduce yields as the pod and seed moisture content decline.
- ▶ Typically, soybean reach harvest grain moisture about 7 to 10 days following full maturity.

SOYBEAN STAGING

Vegetative and reproductive stages are defined as when 50 percent of the field or area of interest has reached a given stage.

VEGETATIVE STAGING

VE – emergence:

the hypocotyl or cotyledons break the soil surface

VC

the unifoliate leaves are unrolled; the unifoliate leaves are the first leaves and develop opposite each other on the main stem

V1

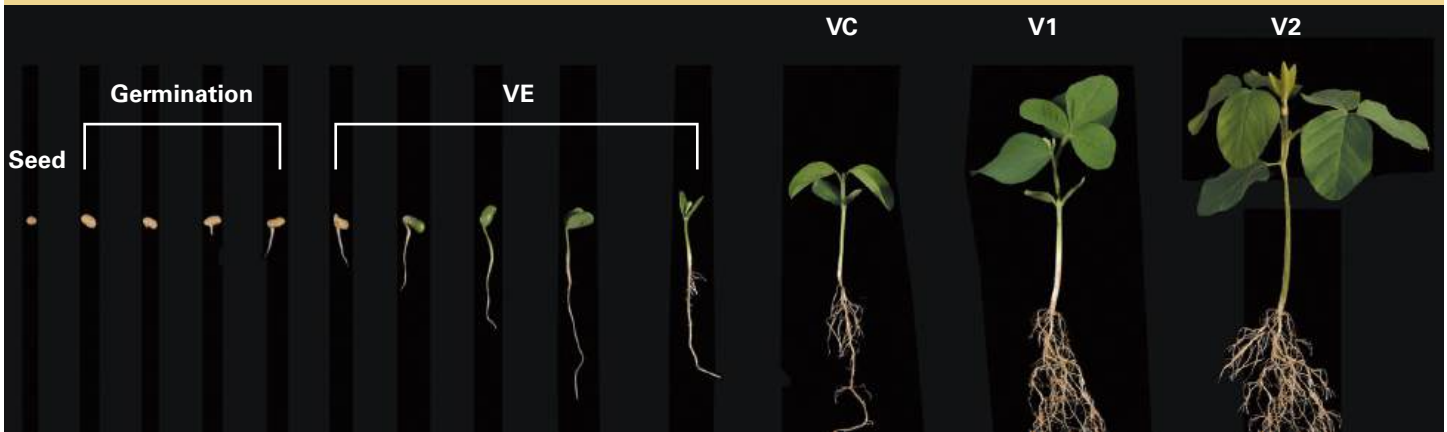
the first trifoliate leaf is unrolled; trifoliate leaves develop singularly and on alternating sides of the stem

V2

the second trifoliate leaf is unrolled

Vn – nth trifoliate leaf:

indeterminate soybean continue to develop trifoliate leaves after reproductive development begins



REPRODUCTIVE STAGING

R1-2 – bloom:

beginning flowering (R1) is when there is one open flower anywhere on the main stem; full flowering (R2) is attained when an open flower is at one of the two uppermost nodes on the main stem

R3-4 – pod development:

beginning pod (R3) is when pods are 3/16-inch long at one of the four uppermost nodes on the main stem; full pod (R4) is when pods are 3/4-inch long at one of the four uppermost nodes on the main stem

R5-6 – seed development:

beginning seed fill (R5) is when seeds are 1/8-inch long at one of the four uppermost nodes on the main stem; full seed (R6) is when the seeds fill the pod cavity at one of the four uppermost nodes on the main stem

R7-8 – maturity:

beginning maturity (R7) is when one pod anywhere on the main stem has reached mature color; full maturity (R8) is when 95 percent of the pods have reached mature color (tan to brown)

